How much of the original distribution is near the sampled vectors?

Measure every original vector's distance to the sample set, then read the shaded histogram area.

1 Sample vectors

Each draw has the same 6 features.

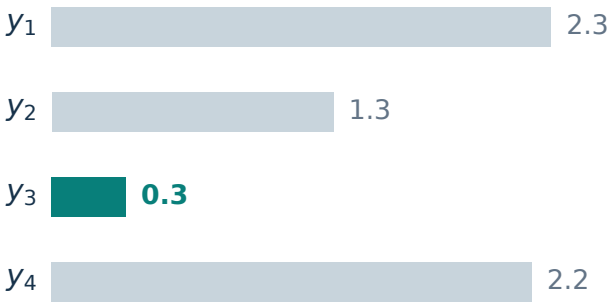
	f1	f2	f3	f4	f5	f6
y_1	+0.4	+0.4	+0.8	+0.4	+1.3	-0.9
y_2	+0.0	-1.0	+0.3	+0.6	+0.3	+0.1
y_3	-0.7	-0.2	-0.2	+1.1	+0.8	+0.4
y_4	+0.4	+0.1	+0.7	-0.2	+0.0	-0.2
x	-0.6	-0.4	-0.0	+1.1	+0.7	+0.5

Example original vector

2 Compute one distance

Find the closest sampled vector.

$$D_C(x) = \min_i d(x, y_i)$$

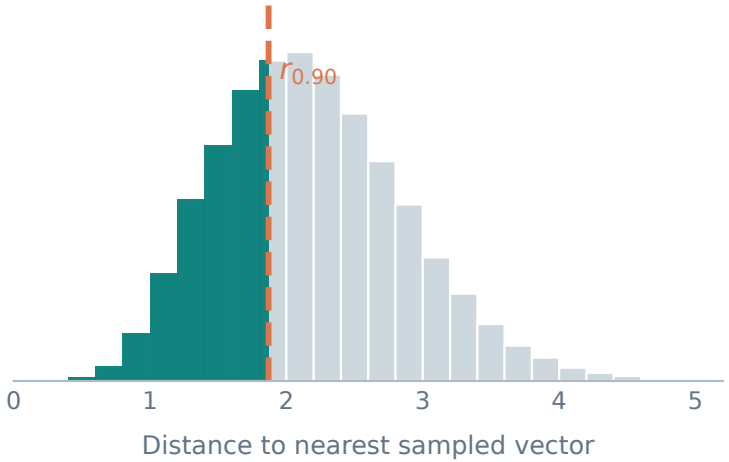


Small distance = close to a sample

Use the full vector, not a single feature.

3 Read the original mass

Histogram distances for original vectors



Shaded mass $A_{0.90} \approx 0.35$

while covering ~90% of sampler draws

Choose the cutoff:

$$r_\alpha = \text{quantile}_\alpha\{D_C(Y^{\text{check}})\}$$

Original probability in that region:

$$A_\alpha = P_0[D_C(X) \le r_\alpha] = \int_{D_C(z) \le r_\alpha} p_0(z)\, dz$$

Illustrative 6D data; α = 0.90